

Equipment awareness

DMMs are available online through electronics distributors, local electronics stores, electrical supply stores, hardware stores, flea markets and online through several e-commerce websites. This easy availability has increased awareness of getting circuits tested before trying to make these operational.

Arindam Majumdar, managing director, Fluke India, division points out, "The realisation that energy saved can be turned into profit is growing with time. Increased awareness to reduce environmental impact, for example, through controlled industrial emissions, is expected to have an impact on the test equipment market."

Major contributors to this report

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up to 7½ digits auto-calibration, it offers signal sampling with 1MS/s, 18-bit digitiser.

What influences multimeter design

Among the factors taken into consideration, Majumdar points out the regulations on the quality of power in the industry and awareness and implementation of electrical safety standards. According to him, "Several industries are now achieving compliance to their industry-specific norms. This, in turn, calls for the need of more accurate, precise and robust T&M technology."

Development of these DMMs requires certain finesse. Since wireless equipment is fundamentally transmitting power in the form of signals, these require a consistent power

source. Manufacturers like Qmax have the option of universal serial bus-powered devices as well, which reduce battery constraint.

Designing also depends highly on requirements, as an engineer would like to utilise as many features as possible. "The most important factor is the customer use case. That will govern the need for supported power levels, clock rates, decoding and so on," says Ben-Dov on the development of DMMs.

DMM: Market and you

"Over the years, T&M equipment has grown in significance, reflecting the critical role it plays in any industry, right from R&D to production and services. Some of the major market drivers include the increasing demand for electronic products and the rising need for T&M equipment for use in semiconductor devices," according to Majumdar.

The need to deliver safe and highly-accurate electronic equip-

ment has led to an increase in employing DMMs during the testing and design process. Singh adds, "CAT ratings, IP ratings, stability and repeatability of readings are important things to decide on a multimeter."

Kumar agrees, "Weighing the pros and cons and considering the use cases are important for proper selection."

What to look forward to

Better precision and support for different integrated protocol/power measurements, according to Ben-Dov, are the future expectations from multimeter devices. "More products will get integrated in the multimeter to reduce the overall cost, and this also allows a technician to carry fewer tools," adds Singh.

Major changes in a multimeter are not the observed pattern. These evolve slowly over the years, and with involvement of wireless capabilities, changes in the near future seem to be confined to the power management system of the device.

Handheld devices generally have a long battery life of about 500 hours in continued usage of AA batteries, making power consumption cost-effective.

Testing of equipment will always be a significant requirement and hence safety of testers will be a responsibility. This is sure to give a boost to the handheld device sector with wireless capabilities. **EFY**

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